

Tamil Nadu Climate Resilient Urban Development Program (TNCRUDP)

Technical Note on Climate Change

Climate Change

Climate Change vulnerability context:

Urban Local Bodies (ULBs) in Tamil Nadu frequently face the combined climate risks of hydro-meteorological hazards (like floods, storm surges, etc.) and slow onset of disasters (like droughts). A review of climate induced risks, specifically water risks, reveals that, among the 21 ULBs participating in the TNCRUDP, 11 face high risk of water scarcity, 6 face high risk to riverine flooding, 3 (namely Kancheepuram, Tambaram and Tiruvarur) face high risk to coastal flooding, while almost all the ULBs face high risk to extreme heat. A further assessment using the Aqueduct Water Risk Atlas indicates that all the ULBs fall under the category of high to extreme water high risk.

Tamil Nadu's vulnerability to landslides is very low to medium (Dindigul, Salem, and Theni have been categorized as medium risk), medium prone to earthquake hazard and only the ULBs along the coast are vulnerable to tsunami¹. Since the risks due to geophysical hazards are very low to medium, they will not have a significant impact on the Water Supply and Sanitation (WSS) projects proposed under the Program. Thus, the table below presents a snapshot of only climate and water risks faced by these ULBs.

ULB-wise climate induced risks and overall water risks

Sl.no	Name of the ULB	Extreme heat, based on think hazard tool	Overall Water Risk (Aqueduct water risk atlas)	Riverine Flood Risk Stress (Aqueduct water risk atlas)
1	Kancheepuram	High	Extremely high	High
2	Pudukottai	High	Extremely high	Extremely high
3	Tambaram	High	Extremely high	Extremely high
4	Salem	High	Extremely high	Low - Medium
5	Dharmapuri	High	Extremely high	Low - Medium
6	Nagercoil	Medium	High	Extremely high
7	Tiruvarur	High	Extremely high	Extremely high
8	Karaikudi	High	Extremely high	Extremely high
9	Dindigul	High	Extremely high	Medium - High
10	Tiruvannamalai	High	Extremely high	Low - Medium
11	Krishnagiri	High	Extremely high	Low - Medium
12	Tiruchirappalli	High	Extremely high	High
13	Avadi	High	Extremely high	Extremely high
14	Vellore	High	Extremely high	High
15	Theni-Allinagaram	Medium	Extremely high	Extremely high
16	Erode	High	Extremely high	Low - Medium
17	Namakkal	High	Extremely high	High
18	Rajapalayam	High	Extremely high	Low - Medium
19	Cuddalore	High	Extremely high	Extremely high
20	Thoothukudi	High	Extremely high	Low - Medium
21	Tirunelveli	High	Extremely high	High

Extreme heat is prevalent in more than 90% of the participating ULBs. Owing to rapid urbanization across the state, unplanned and unsustainable development continues to proliferate at the cost of mass removal of green cover resulting in heat island effect². Urban forests and green cover play a critical role in sustaining watersheds and preventing landslides, especially in hilly terrains like Krishnagiri, regulating a city's microclimate, mitigating heat island effect, and providing carbon sinks for sequestration. While the tree cover could play a critical role in improving the adaptive

¹ According to the Think Hazard tool developed by Global Facility for Disaster Reduction and Recovery (GFDRR) with support from the WBG

² Draft Tamil Nadu State Action Plan on Climate Change - 2.0

capacity of the ULBs by limiting the impact of heat waves, an account of existing green spaces (which include green areas in public and private places, parks, avenue plantations) in the 21 ULBs indicates that only 4 ULBs have more than 1 sq.m. per capita green space (WHO prescribes 9 sq.m. per capita). Furthermore, Urban and Regional Development Plans Formulation and Implementation (URDPFI) guidelines of the Government of India recommend 12% to 18% green cover in urban areas. Except for Krishnagiri which has 1.45% green cover, green cover in none of the other ULBs crosses 1% of total area of the ULB.

More than 48% of urban Tamil Nadu has previously experienced issues with respect to water availability and unsafe sanitation access contributes to high climate-related risks. This is a result of growing demand due to urbanization, poorly managed water systems and climate-induced seasonal extremes. The severe northeast monsoons of 2015 resulted in a record-breaking flood in Tamil Nadu, which was followed by the worst drought in 140 years the state faced in 2016. During the summer of 2019, four major reservoirs experienced complete depletion, leading to a “Day Zero” water crisis in the capital city, Chennai.³ Out of the 1,943 tanks identified by the Ministry of Water Resources, a significant number have been revealed as sites for municipal waste disposal. The fact that **WSS services in TN’s ULBs suffer from limited coverage and quality, as well as high operational and financing inefficiencies exacerbate climate-induced impacts (such as floods and droughts) to the population.** In most ULBs, residents receive water for only one to three hours per day, with the frequency varying from daily to once in every five days. Bulk water supply levels are at 65 liters per capita per day (lpcd) compared to the national guideline of 135 lpcd. Thus, the population often depends on informal water supply (such as water trucks, private vendors or unregulated groundwater) to meet their needs, but these sources tend to become unavailable and unsafe in climate extreme events. The quality of service is further strained by uncontrolled non-revenue water (NRW), which leads to wastage of fresh water sources. **Access to sanitation services is poor as well and raises public health concerns relating to water source contamination from flood and storm waters.** For the one-third of ULBs with sewerage networks, service coverage is less than 40 percent. Many of the existing sewage treatment plants (STPs) suffer from poor operation and maintenance (O&M). A systematic planning through a combination of sewerage and non-sewerage systems to ensure a citywide access to safely managed sanitation is missing. As such, a large part of the population still relies on unsafely managed sanitation, which exposes them to health risks in a flooding event. Women and children also suffer the most from the lack of improved sanitation, which causes diseases and increases the need for health care support, as well as absence from school.

Energy Efficiency is a non-regret, climate-friendly, source of savings to ULBs. The Alliance to Save Energy’s Watergy program launched in 2007 by TNUISL has demonstrated that, by optimizing energy use, municipalities can achieve significant energy and monetary savings, at least 20 percent in water supply systems alone. By updating municipal infrastructure, improved utility services can be provided to the population and industries while simultaneously reducing pollution, GHG emissions and costs. Some of the participating ULBs like Vellore, Cuddalore, Pudukottai and Dindigul were also a part of the municipal energy efficiency project by TNUISL. However, increasing urban population and extreme temperatures will lead to further increase in energy consumptions that attribute to rise in GHG emissions⁴.

ULB’s face institutional capacity and policy challenges. Insights from consultations with ULBs revealed that tackling climate change is presently limited to adaptation measures like preparedness for disaster management. The latest government order with sanctioned positions for ULBs in Tamil Nadu does not contain dedicated personnel for climate change and resilience, thus limiting planning, financing and rolling out robust climate actions. Furthermore, ULBs currently lack mechanisms to incorporate climate and disaster risks in their investment identification and prioritization procedures, and there is a lack of contingency planning for impending risk of climate hazards. Thus, there is a strong case for enhancing the institutional capacities and increasing investments for incorporating climate resilience in ULBs and the proposed urban WSS infrastructure systems.

Green House Gases accounting:

An estimation of emissions from wastewater projects was carried out for the participating ULBs. The study shows that without the implementation of proposed fourteen sewerage projects under the Program, the GHG emissions will continue to increase owing to the CH₄ (methane)

³ <https://thecityfix.com/blog/tamil-nadus-seasonal-swings-water-stress-state-much-little-water/>

⁴ [Ministry of Power – Tamil Nadu](#)

emissions from untreated wastewater. GHG account was carried out based on the Water Global practice standardized methodology, focusing on sanitation activities (as the most relevant sector on this matter). It shows that the Program will result in a reduction in unchecked GHG emissions from individual cesspools and latrines by expanding household sewerage connections and increasing the amount of wastewater being treated in existing and newly constructed STPs in the project area that meet the effluent disposal standards. The Program's gross emissions over the economic lifetime are **2,318,677 tCO₂-eq**, the net emission reduction over the economic lifetime is **6,786,713 tCO₂-eq**, and the net average annual emission reduction is **212,085 tCO₂-eq**.⁵

In the WSS infrastructure assets financed under the Program, energy consumption from pumping stations, motors, and treatment plants are potential sources of GHG emissions due to expansion of WSS services in the project area. However, these will be offset by emissions reductions due to more households being connected to sewerage solutions, moving away from use of highly emissive septic tanks and also by expected reduction in emissions due to energy efficiency gains (see DLI 8), non-revenue water reduction (DLI 7.2) and creation of green areas (DLI 8). In the medium to long term, further reduction in emissions due to progressive city-wide access to safely managed sanitation solutions coming from the implementation of the CWIS plans (DLI 3) are expected. These aspects (energy efficiency, parks, NRW, CWIS) have not been quantified in the GHG calculations.

Disbursement-Linked Indicator (DLI)-wise Climate Adaptation and Mitigation Actions

Activities	Adaptation Action	Mitigation Action
Results Area 1: Strengthening urban management, institutional framework and climate resilient planning		
DLI 2: Increased access to financial market		
ULBs issue municipal bonds with support from TNUIFSL	Funds raised through these bonds will be allocated to the ULBs for investing in urban water systems including WSS and stormwater drainage.	
DLI 3: Enhanced institutional capacities for climate resilient planning		
Adoption and initiation of implementation of climate change action plans by participating ULBs		
• ULBs have prepared in a consultative manner, adopted and initiated implementation of City-Climate Action Plan (CCAP) • The CCAP should include a cross-sectoral action plan covering WSS, transport, energy/building, solid waste and green space, and four WSS sub-plans on water security and emergency preparedness and response, citywide inclusive sanitation (CWIS), WSS business plan, and stormwater drainage.	<i>Vulnerability context:</i> Climate risks as described above. <i>Intent to address identified vulnerabilities:</i> This DLI and DLRs will significantly contribute to adaptation efforts by strengthening the institutional capacity of ULBs to tackle impacts of climate change including the extreme weather events like floods, droughts, cyclones, etc. <i>Explicit link between identified climate change risks and specific project activities:</i> The cross-sectoral CCAP and four WSS subplans will include recommendations that prioritizes: (i) water source sustainability, reduction of NRW to also contribute to water security, installation of rainwater harvesting systems. All of this address the risks of water scarcity in the participating ULBs. The emergency preparedness and response plan, enhances the capacity of the ULBs, to better handle climate crisis. This would ensure that the WSS infrastructure are resilient, and the service delivery is not disrupted during	The CCAPs for the ULBs will comprise of mitigation actions for reducing GHG emissions, by establishing set emission reduction targets and have planned approach towards decarbonization. The CWIS plan support setting the ULBs on a progressive path to emissions reduction by increasing access to safely managed sanitation services, including increase the treatment of sewerage, leading to reduction of CH₄, CO₂ and N₂O emissions.

⁵GHG account was carried out based on the Water Global practice standardized methodology, focusing on sanitation activities only since these are the largest portion of program-supported investments with relevance to GHG accounting. DLIs supporting sanitation investments are: DLIs 6.1 (partially), 6.3 (fully), 7.1 (partially) and 7.3 (fully).

Activities	Adaptation Action	Mitigation Action
	disasters, which would further reduce the stress on water resources. (ii) urban drainage plan to identify hot spots of water stagnation and recommend structural and non-structural measures including nature-based solutions to improve the stormwater management system to reduce flooding damage; (iii) CWIS plan to support ULBs to progressively advance in city-wide access to improved and safely managed sanitation, thus supporting reducing health impacts in a flooding event.	
Results Area 2: Improving accessibility, climate resilience and efficiency of urban WSS services		
DLI 6: Delivery of functional water and sanitation connections		
DLI 6.1: ULBs implementation of WSS investments incorporating climate resilience guidelines		
DLI 6.2: Number of new or rehabilitated functional water tap connections		
DLI 6.3: Number of new functional sewerage connections		
- WSS Investment implementation by participating ULBs incorporating climate resilience guidelines developed by TNUIFSL - Number of new or rehabilitated functional water tap connections across 2 ULBs with pan-city of city water supply projects and 12 ULBs with 24x7 water supply projects of the city - Number of new functional sewerage connections (treated effluents up to Effluent Discharge Standard levels) in 14 ULBs	<i>Vulnerability context:</i> Water risks and climate risks discussed above. <i>Intent to address identified vulnerabilities:</i> Development of climate resilient infrastructure for service delivery and operations. <i>Explicit link between identified climate change risks and specific project activities:</i> DLIs 6.2 and 6.3 mandate the incorporation of climate resilience measures in the design of the WSS investments funded under the Program. These include measures to prevent potential damages due to floods, wind and droughts and other climatic hazards. Increasing the number of functional tap connections is a more climate resilient water supply solution that reduces the use of unsafe water sources, especially during 'stress' scenarios of flooding and cyclone, as informal water points can be difficult to access or become contaminated. Access to functional sewerage connections refers to the entire sanitation chain (from household connection up to treating the effluent as per standards) thus enhancing the resilience of infrastructure and services to climate risks and directly contributing to reduce health risks exposures in floods events.	The implementation of wastewater projects under the Program will lead to reduction of CO ₂ emissions (see GHG paragraph above). Increased household sewerage connections in addition to augmentation of existing network, as well as new STPs will contribute to increase in treatment quantity of sewerage generated in the ULBs.
DLI 7: Operational, climate and financial efficiency of WSS services		
DLI 7.1: Number of participating ULBs with performance-based contracts signed for water and/ or sanitation services		
DLI 7.2: Number of water connections with NRW 20% or below in areas receiving 24 x 7 services		
- Number of ULBs that have signed performance-based contracts for water and/ or sewerage services. - Number of ULBs with Non-Revenue Water 20% or below	<i>Vulnerability context:</i> Water risks discussed above. <i>Intent to address identified vulnerabilities:</i> Ensuring resilient and well operated WSS systems at a ULB level. Reduced water losses and improved water and wastewater management.	

Activities	Adaptation Action	Mitigation Action
(measured for at least six months) in project areas up to household level in 24x7 projects and [up to DMA level] in citywide water projects	<i>Explicit link between identified climate change risks and specific project activities:</i> PBC contracts will finance implementation and operation and maintenance of WSS services, as per DLI 6 and additional WSS efficiency improvements as per DLIs 7.2 and 7.3. Specific clauses under the performance-based contracts will ensure, contractors are incorporating the necessary mitigation measures, that would contribute to building resilience for the WSS systems in the ULBs. The projects for water and sewerage are embedding climate resilience in the project design as prescribed under DLI 6.1, and the PBCs also includes the portion of O&M of these WSS services. Reduction of NRW, further reduces stress on freshwater resources.	
DLI 7.3 Increase in cumulative volume of sewage treated (meeting minimum standards) in eligible existing sewage treatment plants (STPs)		
Increase in volume of treated wastewater in existing STPs	DLI 7.3 supports higher sewage treatment, thus reducing health impacts in case of floods.	Increase in treatment of wastewater in existing STPs will reduce greenhouse gas emissions during the Program period.
DLI 8: Improved city-level climate resilience		
DLI 8.1: Energy savings over respective ULB baseline of FY2021-22		
ULBs have demonstrated aggregate energy savings.		Implementation of energy efficiency measures in municipal assets ensures progressive reduction in emissions.
DLI 8.2: Increase in area of green space		
ULBs' have greened more area designated as parks	Increasing green space will help mitigate heat island effects and increase rainfall infiltration, thus reducing flood impacts.	Increasing green space will contribute to carbon sequestration.
Program Action Plan		
Establish water and wastewater quality surveillance program and disclose results		
Within 3 years of Program implementation, ULBs will establish the surveillance program for city-wide sampling, testing and disclosure of water quality and wastewater influent and effluent quality as per applicable norms, and disclose results on a quarterly basis	Vulnerability context: Climate risks discussed above. Disclosure of water quality and wastewater quality reports will ensure periodic monitoring of water and effluents in a systematic manner, thus incentivizing ULBs to keep up the quality of the services and their climate resilience function.	Disclosure of wastewater quality reports will ensure periodic monitoring of sewerage in a systematic manner. This will lead to ULBs to complying with the Effluent Discharge standards of Pollution Control Board with reduction in BOD levels contributing to emissions reduction.

Paris Alignment

The operation is aligned with the goals of the Paris Agreement focused on mitigation, adaptation, and resilience.

Consistency with the country's NDC, LTS, and NAP A review of the wastewater management projects under the Program shows a strong potential to reduce GHG emissions as the projects will lead to reducing methane emissions from untreated wastewater through scientific treatment and disposal of the same (see GHG accounting paragraph). Moreover, low carbon technological design and interventions and the City Climate Action Plan (CCAP) (such as energy efficiency, increased treatment of the wastewater, CWIS, green areas) will ensure future WSS projects are set on a low

carbon trajectory, contributing to the overall climate goals of the state (in line with INDCs). To ensure that all Program activities are aligned with India's NDCs, India's Long-Term Low-Carbon Development Strategy (2022), and National Action Plan on Climate Change (NAPCC 2021), specific DLRs have been proposed (DLRs 3 and 8, and parts of DLR 6 and 7) which include clear activities and results towards climate adaptation and mitigation.

Climate Baseline assessment: climate vulnerability situation of program ULBs was described earlier in this document. On the institutional responses side, it was identified that tackling climate change is limited preparedness for disaster management. ULBs in Tamil Nadu does not contain dedicated personnel for climate change and resilience, thus limiting planning, financing, and rolling out robust climate actions. Furthermore, ULBs currently lack mechanisms to incorporate climate and disaster risks in their investment identification and prioritization procedures, and there is a lack of contingency planning for impending risk of climate hazards.

Climate resilience interventions are integrated under both the result areas under the Program.

While RA1 focuses on strengthening urban management and climate resilient planning at ULB level, RA2 prioritizes improving accessibility, climate resilience and financial sustainability of urban services. On one hand, RA1 prioritizes the adoption of CCAP for each ULB, which includes climate risk vulnerability assessment, GHG inventories, adaptation and mitigation measures based on their vulnerabilities, and identifying alternative water sources to ensure water security. On the other hand, RA2 incentivizes the ULBs to mainstream climate resilience aspects in investment design and O&M of proposed WSS infrastructure, including adopting a detailed climate risk screening checklist and incorporating resilient measures in the design of their WSS projects. The Program incentivizes the development and approval of the water security and emergency plan for WSS services (RA1), which are intended to better prepare the service providers to maintain WSS services (RA2) in case of a climate-related emergency. The use of a program for results instrument (PforR) supports carrying out climate related activities in different levels: physical designs, O&M of assets, service delivery, planning, institutional strengthening.

Assessment and reduction of mitigation risks: Program has a low risk of preventing the country's transition to low-carbon development pathways, since (i) all the activities financed are either universally aligned or are expected to lead to low GHG emission impacts; (ii) a mechanism for tracking and lowering GHG emissions for activities with moderate GHG emissions intensity have been assessed and integrated in the Program design; (iii) activities supported by the Program and the economic impacts of the PforR are not expected to be at risk from low-carbon transition. The operation is not at material risk of having a negative impact on the country's low-GHG emissions development pathways. All of the activities financed under the Program are on the universally aligned list, including the expansion and rehabilitation of water supply systems⁶, NRW management, wastewater management systems (from collection to effluents treatment), conducting of energy audits and improving of energy efficiency, and constructing of green spaces.

The proposed wastewater projects represent the largest portion of program investments. In this sector, Program activities support a full sanitation chain, from household connections all the way through treating and discharging effluents at adequate standards. These activities enable a robust reduction in GHG emissions, as described earlier in this document. For the WSS infrastructure assets financed under the Program, energy consumption from pumping stations, motors, and treatment plants are the potential sources of GHG emissions. To ensure that all Program activities are aligned with India's NDCs, India's Long-Term Low-Carbon Development Strategy (2022), and NAPCC 2021, specific targets have been proposed under RAs 1 and 2. The Program will incentivize the adoption of robust CCAP by each ULB, which includes preparation of CWIS plans to improve sanitation facilities and city-wide safe disposal of sewage, leading to a progressive (medium-to-long term) reduction of GHG emissions. Increasing green space will help mitigate heat island effects, increase rainfall infiltration, and contribute to carbon sequestration. The ULBs are incentivized to carry out energy audits to closely monitor their energy consumption and achieve systematic improvement in energy efficiency for their municipal assets. The Program will also ensure progressive reduction in emissions through incentivizing the optimization of existing assets and efficient operation of WSS assets. The proposed wastewater management subprojects would lead to an increase in sewerage connections and treatment of wastewater, directly leading to a net reduction in lifetime emissions of 6,786,713 tCO₂-eq.

⁶ All WSS pumping systems are supplied by the grid. However, occasionally in events of power failure, some critical pumps may be supplied by diesel generators on a temporary basis.

Assessment and reduction of adaptation risks. The ULBs in Program areas have a high exposure to climate change risks, particularly water scarcity, floods, and extreme heat. The Program's climate risk reduction measures would help the 21 participating ULBs to adapt to them. RA1 prioritizes the adoption of CCAPs for the ULBs based on assessment of climate vulnerability and GHG inventories, and implementation of adaptation and mitigation measures including identifying alternative water sources to ensure water security, promoting city-wide access to improved sanitation (including adequate treatment) and ensuring water supply and sanitation infrastructure are resilient to flooding and other disasters. RA2 incentivizes the ULBs to mainstream climate resilience aspects in investment design, planning, and operation of WSS services. A climate risk screening checklist and resilience design guideline have been developed by TNUIFSL outlining climate risk aspects and resilience measures pertaining to WSS systems, and the ULBs are required to incorporate these measures in the design of their WSS projects. This will help pre-empt and prevent potential impacts to WSS systems from climate-exacerbated floods and droughts, for example, restricting potential contamination of raw water in service reservoirs due to flooding, reducing disruption in road connectivity, and ensuring access to pump houses during flooding. In the case of wastewater management systems, enhanced resilience to floods will lead to lower operational disruptions caused by power fluctuations causing obstructions in pumping, which may lead to sewage overflows and flooding of STPs, while drought preparedness will limit the impact of silting of sewers due to low flows. For each type of risk, the guideline recommends specific measures for the design and O&M stages.

Overall, core Program activities such as improvement in WSS coverage, performance standards, and an overall effective institutional framework will enhance operational efficiency and sustainability of services, thus helping enhance resilience to droughts, floods, and extreme temperatures. The IPF component will also include training and capacity building for the ULBs in the management of WSS services during climate emergencies. In addition, through DLI 8, ULBs are encouraged to adopt a targeted approach to achieve systematic improvement in energy efficiency for municipal assets. ULBs will also achieve increase in their carbon sink, through developments of parks which will lead to increase in green cover. Apart from carbon sequestration benefits, this will also have significant adaptation benefits by preventing the formation of urban heat islands. Disbursement linked to these indicators, aims to serve as necessary incentive for ULBs to prioritize improved climate performance through progressive implementation of specific and relevant climate actions. Thus, risks from climate hazards have been reduced to an acceptable level and are not likely to have material impact on the operation and its development objective.